

Task 2A: PostgreSQL OLAP Schema (Analytical)

File Name: Part 2_Task 2A.sql

This script implements a **Star Schema** to optimize the system for healthcare analytics and reporting.

SQL

```
-- PART 2 TASK 2A: OLAP SCHEMA (PostgreSQL)
-- DIMENSIONS
CREATE TABLE Dim_Patient (PatientSK SERIAL PRIMARY KEY, PatientID INT, FullName
VARCHAR(100), DOB DATE);
CREATE TABLE Dim_Employee (EmployeeSK SERIAL PRIMARY KEY, EmployeeID INT, FullName
VARCHAR(100), RoleName VARCHAR(50));

-- FACT TABLE
CREATE TABLE Fact_Healthcare_Analytics (
    FactID SERIAL PRIMARY KEY,
    PatientSK INT REFERENCES Dim_Patient(PatientSK),
    EmployeeSK INT REFERENCES Dim_Employee(EmployeeSK),
    BillingAmount DECIMAL(12,2),
    VisitCount INT DEFAULT 1
);

-- ETL MOCK INSERT
INSERT INTO Dim_Patient (PatientID, FullName, DOB) VALUES (101, 'John Doe', '1985-06-15');
INSERT INTO Dim_Employee (EmployeeID, FullName, RoleName) VALUES (501, 'Dr. Smith',
'Doctor');
INSERT INTO Fact_Healthcare_Analytics (PatientSK, EmployeeSK, BillingAmount) VALUES (1, 1,
350.00);
```